

**BASAVARAJESWARI GROUP OF INSTITUTIONS
BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT**

NBA and NAAC Accredited Institution*

(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to Visvesvaraya Technological University, Belagavi) "JnanaGangotri" Campus, No.873/2,
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**DEPARTMENT OF
CSE-ARTIFICIAL INTELLIGENCE**

**A
Internship Report
On**

**“ Cloud Computing Internship Report on
Amazon Web Services (AWS)”**

A dissertation submitted to the Department of CSE-Artificial Intelligence of
Visvesvaraya Technological University in partial fulfillment for the Degree of
Bachelor of Engineering award.

Submitted By

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3BR22CA019

**Under the Guidance of
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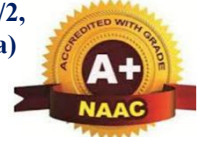


**Visvesvaraya Technological University
Belagavi, Karnataka
2025-2026**

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CERTIFICATE

Certified that the project work entitled “**Cloud Computing Internship Report on Amazon Web Services (AWS)**” carried out by **Ms. Hemalatha D** bearing USN **3BR22CA019** bonafide student of Ballari Institute of Technology and Management in partial fulfillment for the award of Bachelor of Engineering in CSE-Artificial Intelligence of the Visvesvaraya Technological University, Belgaum during the year 2025-2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The internship project report has been approved as it satisfies the academic requirements concerning the internship project work prescribed for the Degree.

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EXTERNAL VIVA

Name of the Examiners
1.
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Signature and date
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ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of internship project work on “**Cloud Computing Internship Report on Amazon Web Services (AWS)**” would be incomplete without the mention of the people who made it possible, whose noble gesture, affection, guidance, encouragement, and support crowned our efforts with success. It is our privilege to express the gratitude and respect to all those who inspired us in the completion of this internship project work.

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HEMALATHA D

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ABSTRACT

This report summarizes the learning experience gained during a 90-day internship in Cloud Computing using Amazon Web Services (AWS). The internship focused on understanding core cloud concepts and gaining practical knowledge of AWS services. Key areas covered include compute, storage, networking, databases, security, monitoring, and cost management. Services such as EC2, S3, RDS, VPC, IAM, and CloudWatch were explored to understand their real-world applications. Overall, the internship helped in building a strong foundation in cloud computing, improving technical skills, and preparing for future opportunities in the cloud domain.

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CHAPTER 1

COMPANY PROFILE

1.1 Company Name: Smartbridge Educational Services Pvt Ltd

SmartBridge Educational Services Pvt Ltd is an EdTech company established in 2015, focused on bridging the gap between academia and industry. It provides industry-relevant training and practical exposure to help students become job-ready professionals.

Through its platform “SmartInternz,” the company offers project-based learning and virtual internships, enabling students to develop technical and professional skills. SmartBridge has trained and upskilled a large number of students and faculty in emerging technologies.

It mainly focuses on:

- Skill Development & Training
- Virtual Internships
- Industry-Academia Collaboration

1.2 Company Address

TELANGANA: 6th floor, Sundarayya Vignana Kendram, Technical Block, Madhava Reddy Colony, Gachibowli, Hyderabad-500032

NOIDA: 3rd Floor, Big Box Co-working Space, Office no 303, Pinnacle Tower - 142, Plot no. 8, Sector 142, Noida, Uttar Pradesh 201301

1.3 Phone Number

+91 087120 34656

1.4 Website URL

<https://www.thesmartbridge.com/>

1.5 Email Address

info@thesmartbridge.com

1.6 Major Organization Milestones

1.6.1 Transforming Education with Scalable Learning

At SmartBridge Educational Services Pvt Ltd, we are committed to transforming education at scale by bridging the gap between academia and industry. Through our structured programs and collaborations with institutions, we ensure that students gain industry-relevant skills and practical exposure. Our approach focuses on delivering measurable impact across learners, educational institutions, and corporate partners, empowering students to become job-ready professionals.

1.6.2 SmartInternz Platform for Skill Development

We understand the importance of practical learning in today's competitive environment. Our innovative platform, SmartInternz, provides project-based and collaborative learning experiences integrated into academic curricula. This enables students to develop both technical and professional skills required by the industry. The platform ensures flexibility, accessibility, and hands-on experience, helping students learn effectively and stay ahead in their careers.

1.6.3 Industry Collaboration and Virtual Internships

At SmartBridge, we strongly believe in industry-academia collaboration. By partnering with leading organizations, we provide students with valuable virtual internship opportunities in emerging technologies. Our initiative to deliver "1 Million Virtual Internships" aims to equip students with real-world experience and enhance their employability, while also helping organizations find skilled and job-ready talent.

1.6.4 Building a Skilled Workforce for the Future

Our mission is to build a sustainable talent pipeline by equipping students with in-demand skills through structured training programs. With a strong focus on innovation, equal opportunities, and outcome-driven learning, SmartBridge has successfully upskilled millions of learners. We continue to contribute towards workforce development and support the growth of a skilled and future-ready generation.

CHAPTER 2

ABOUT THE COMPANY

2.1 History of the Organization

Founding (2015): SmartBridge Educational Services Pvt Ltd was founded in 2015 with the objective of bridging the gap between academia and industry. The organization was established by professionals who identified the need for industry-ready skills among students and aimed to provide practical learning solutions aligned with current market requirements.

Initial Focus (2015-2018): In its early years, SmartBridge focused on providing skill development programs and training initiatives for students. The organization collaborated with educational institutions to introduce industry-oriented learning methods and enhance the employability of students.

Expansion and Program Development (2018-2020): As the demand for skilled professionals increased, SmartBridge expanded its operations by partnering with multiple colleges and universities. It developed structured training programs based on emerging technologies and industry standards, ensuring that students gained relevant technical knowledge and hands-on experience.

SmartInternz Platform Launch (2020): A major milestone in the organization's journey was the launch of the SmartInternz platform. This platform provided project-based learning and virtual internship opportunities, enabling students to gain practical exposure and real-world experience regardless of their location.

Growth and Industry Collaboration (2020–Present): SmartBridge strengthened its presence by collaborating with leading organizations such as IBM, Google, Salesforce, and VMware. Through these partnerships, it provided large-scale virtual internships and industry-recognized training programs, benefiting a wide range of students.

Global Impact and Future Vision (Present): Today, SmartBridge has upskilled millions of learners and continues to expand its reach across institutions and industries. The

organization is working towards its mission of providing “1 Million Virtual Internships” and building a strong, future-ready workforce.

2.2 Organization Structure

2.2.1 Executive Leadership

Founder: Oversees the overall vision, strategy, and growth of SmartBridge.

Co-Founder: Supports strategic planning and development of training programs.

Technology Head: Manages development and maintenance of learning platforms.

Operations Head: Handles daily operations and execution of programs.

Human Resources (HR): Manages recruitment, coordination, and organizational activities.

2.2.2 Departments

Training and Development: Responsible for designing and delivering skill-based training programs aligned with industry needs.

Technology Department: Manages platform development, maintenance, and innovation of digital learning systems.

Partnership and Placement: Handles collaborations with educational institutions and corporate partners, and supports internship opportunities.

Content Development: Develops course materials, projects, and assessments for effective learning.

Operations and Support: Ensures smooth execution of programs, student coordination, and overall administrative support.

2.3 Products And Services Offered

SmartInternz: A project-based learning and remote internship platform that provides industry-partnered virtual internships. It offers hands-on experience in domains such as AI/ML, Cloud Computing, Cybersecurity, and IoT through real-time projects.

Skill Wallet: A career readiness platform that allows students to track their skills, earn badges, and maintain a digital portfolio. It helps in measuring and showcasing their “Skill Network” to improve employability.

Center for Future Skills & Innovation (CFSI): An on-campus model designed for educational institutions to promote industry-aligned skill development and project-based learning within the curriculum.

Virtual Internship Programs (VIP): Structured internship programs conducted in collaboration with leading companies like IBM, Google, Salesforce, and VMware, providing practical exposure through real-world projects.

Industry-Aligned Credit Courses: Courses integrated into academic curriculum as professional electives, combining technical training with project development.

Faculty Development Programs (FDP): Training programs for faculty members focusing on modern teaching methodologies and emerging technologies.

LastMile Program: Job-readiness program that focuses on developing role-specific skills, interview preparation, and career guidance.

Hackathons and Innovation Challenges: Events conducted to encourage problem-solving skills, innovation, and practical implementation of knowledge.

CSR Program Execution: Implementation of corporate-sponsored skill development programs such as IBM SkillsBuild to enhance student capabilities.

2.4 Employee Structure of the Organization

Directors: The organization is led by directors Amarender Katkam and Suman Akula, who are responsible for strategic decisions, overall growth, and maintaining strong industry-academia collaboration.

Program Managers: Oversee virtual internship programs and project-based learning activities. They manage execution, coordinate with institutions, and ensure smooth delivery of training programs.

Technical Mentors/Trainers: Subject matter experts who provide technical guidance in domains such as AI/ML, Data Science, Cloud Computing, and IoT. They support students through training sessions and project mentoring.

Technical Support & QA Team: Responsible for maintaining platform functionality, resolving technical issues, and ensuring quality and performance of learning systems.

Operations & HR Team: Handles recruitment, student coordination, internal operations, and partnerships with academic institutions and corporate organizations.

2.5 Techniques Applied in the Company

Project-Based Learning (PBL): SmartBridge follows a hands-on learning approach where students gain knowledge by working on real-time projects instead of only theoretical concepts. This helps in improving practical skills and problem-solving abilities.

Virtual Internship Model: The organization provides remote internship programs in collaboration with industry partners such as IBM, Google, and Salesforce. These internships offer real-world experience through structured project work.

Industry-Aligned Training: Training programs are designed in collaboration with leading technology companies to match current industry requirements and job roles, ensuring students become job-ready.

Skill-Integrated Learning: SmartBridge integrates technical training with project work, allowing students to earn academic credits while gaining practical exposure.

Hackathons and Innovation Challenges: The company conducts hackathons and competitions to encourage innovation, teamwork, and rapid problem-solving skills among students.

AI-Based Career Guidance: Through platforms like Skill Wallet, SmartBridge provides personalized learning paths, skill tracking, and career recommendations to students.

Faculty Development Programs: The organization trains educators on modern technologies and teaching methodologies to ensure effective knowledge delivery.

Cloud-Based Learning Platforms: Platforms like SmartInternz enable scalable, flexible, and remote learning, allowing students to access training and internships from anywhere.

2.6 Process Adopted

2.6.1 Planning and Requirement Analysis

Identify Skill Gaps: SmartBridge identifies the gap between academic learning and industry requirements through research and collaboration with industry partners.

Define Objectives: Sets clear goals for industry-relevant training and employability.

Curriculum Alignment: Courses and internship programs are designed based on current technologies such as AI/ML, Cloud Computing, and IoT.

2.6.2 Design and Development

Project-Based Learning Design: Training modules are structured around real-world projects using platforms like SmartInternz.

Platform Integration: Learning is delivered through digital platforms such as SmartInternz and Skill Wallet for tracking student progress.

Industry Collaboration: Curriculum and projects are co-designed with industry experts to ensure relevance.

2.6.3 Implementation and Training

Virtual Internships: Students participate in structured virtual internship programs with hands-on project experience.

Skill Development: Technical training is provided in emerging technologies along with mentoring.

Faculty and Mentor Support: Continuous guidance is given by mentors and trainers throughout the learning process.

2.6.4 Evaluation and Assessment

Performance Evaluation: Students are assessed through assignments, projects, and practical-tasks.

Skill Tracking: Platforms like Skill Wallet are used to monitor progress and showcase skills.

Feedback Mechanism: Regular feedback is provided to improve learning outcomes.

2.6.5 Certification and Career Support

Certification: Students receive certificates upon successful completion of training and internships.

Career Readiness: Programs like LastMile focus on interview preparation and job readiness.

Industry Integration: Students are connected with companies through partnerships, job fairs, and hiring programs.

2.6.6 Continuous Improvement and Expansion

Program Enhancement: Courses are regularly updated to align with industry trends such as Generative AI, Cloud Computing, and Data Science, ensuring students gain relevant technical skills.

Strategic Partnerships: The organization strengthens collaborations with leading companies like Salesforce, AWS, and ServiceNow to provide industry-recognized training, certifications, and real-world exposure to students.

Global Expansion: The organization is expanding its reach to international markets.

CHAPTER 3

OBJECTIVES OF INTERNSHIP

3.1 Introduction

The primary objective of this internship was to gain a comprehensive understanding of cloud computing and acquire practical knowledge of Amazon Web Services (AWS). The internship aimed to bridge the gap between academic learning and real-world industry requirements by providing hands-on experience with cloud technologies.

3.2 Specific Objectives

3.2.1 Understanding Cloud Computing Fundamentals

To understand the basic concepts of cloud computing, including service models such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS), along with deployment models like public, private, and hybrid cloud.

3.2.2 Learning Core AWS Services

To gain knowledge and practical experience with core AWS services such as Amazon EC2 for compute, Amazon S3 for storage, and Amazon RDS for database management, which are essential for building cloud-based applications.

3.2.3 Understanding Cloud Networking

To learn networking concepts in AWS using Amazon VPC, including subnets, route tables, internet gateways, and security groups for designing secure and scalable network architectures.

3.2.4 Implementing Cloud Security Practices

To understand security mechanisms such as Identity and Access Management (IAM), roles, policies, multi-factor authentication, and data encryption to ensure secure access and data protection.

3.2.5 Monitoring and Resource Management

To learn how to monitor and manage cloud resources using AWS tools like CloudWatch and CloudTrail for tracking performance and maintaining system reliability.

3.2.6 Cost Management and Optimization

To understand AWS pricing models and use tools like Cost Explorer and Budgets to monitor, control, and optimize cloud costs effectively.

3.2.7 Enhancing Problem-Solving Skills

To develop analytical thinking and problem-solving skills through scenario-based learning and real-world examples in cloud computing.

3.2.8 Certification Preparation

To prepare for the AWS Certified Cloud Practitioner (CLF-C02) certification by strengthening conceptual understanding and practical knowledge.

3.3 Summary

Overall, the internship aimed to provide a balanced combination of theoretical knowledge and practical experience, enabling the development of skills required to work in cloud computing environments.

CHAPTER 4

LEARNING EXPERIENCES

4.1 Introduction

The internship provided a comprehensive learning experience in cloud computing using Amazon Web Services (AWS). It was structured in a systematic manner, starting from basic concepts and progressing towards advanced topics, which helped in building both theoretical knowledge and practical skills.

4.2 Learning Phases

4.2.1 Cloud Computing Fundamentals

In the initial phase, I learned the fundamentals of cloud computing, including its advantages over traditional systems, service models (IaaS, PaaS, SaaS), and deployment models (public, private, hybrid). I also understood AWS global infrastructure such as regions and availability zones.

4.2.2 Compute Services (EC2 & Scaling)

I gained hands-on knowledge of Amazon EC2, where I learned how to launch, configure, and manage virtual machines. I also studied Auto Scaling and Elastic Load Balancing, which help in maintaining application performance and availability.

4.2.3 Storage Services (S3, EBS, EFS)

I explored different storage services such as Amazon S3 for object storage, EBS for block storage, and EFS for file storage. I learned how to choose appropriate storage based on application requirements and optimize storage costs.

4.2.4 Database Services

I learned about database services including Amazon RDS, DynamoDB, and Redshift. This helped me understand how to manage structured and unstructured data efficiently in cloud environments.

4.2.5 Networking (VPC)

I studied networking concepts using Amazon VPC, including subnets, route tables, internet gateways, and NAT gateways. I also learned how to design secure and scalable network architectures.

4.2.6 Security and IAM

Security was a major part of the internship. I learned about IAM users, roles, policies, and multi-factor authentication. I also understood encryption and security best practices in

AWS.

4.2.7 Monitoring and Management Tools

I gained experience with monitoring tools such as AWS CloudWatch and CloudTrail. These tools helped in tracking system performance, logging activities, and maintaining system reliability.

4.2.8 Cost Management and Optimization

I learned about AWS pricing models and cost management tools such as Cost Explorer and Budgets. This helped me understand how to monitor and optimize cloud expenses effectively.

4.2.9 Practice and Assessment

Towards the end of the internship, I practiced mock tests and scenario-based questions. This helped in improving my understanding and preparing for AWS certification.

CHAPTER 5

LEARNING OUTCOMES

5.1 Introduction

The AWS internship provided a strong foundation in cloud computing by combining theoretical understanding with practical exposure. It enhanced my technical knowledge and prepared me to work with real-world cloud-based systems.

5.2 Key Learning Outcomes

5.2.1 Strong Foundation in Cloud Computing

Developed a clear understanding of cloud computing concepts, including service models (IaaS, PaaS, SaaS) and deployment models. This helped in understanding how cloud services are designed and delivered.

5.2.2 Practical Knowledge of AWS Services

Gained hands-on experience with core AWS services such as EC2, S3, and RDS. Learned how to create, configure, and manage cloud resources effectively.

5.2.3 Understanding of Networking and Security

Acquired knowledge of networking concepts using VPC, subnets, and routing. Also understood security practices through IAM, access control, and data protection techniques.

5.2.4 Monitoring and Cost Optimization Skills

Learned to monitor system performance using CloudWatch and CloudTrail. Developed the ability to manage and optimize cloud costs using AWS pricing tools and best practices.

5.2.5 Improved Problem-Solving and Career Readiness

Enhanced analytical thinking and problem-solving skills through practical exercises and scenario-based learning. The internship also improved confidence and readiness for AWS certification and future career opportunities.

CONCLUSION

The 90-day internship in Amazon Web Services (AWS) has been a highly valuable and enriching experience. It provided a comprehensive understanding of cloud computing concepts along with practical exposure to real-world applications. Throughout the internship, I gained knowledge of various AWS services such as EC2, S3, RDS, VPC, IAM, and CloudWatch, which helped me understand how cloud-based systems are designed and managed.

The internship enhanced my technical skills in areas such as cloud architecture, networking, security, and resource management. I also learned how to monitor system performance and optimize costs using AWS tools and best practices. The hands-on exercises and scenario-based learning significantly improved my problem-solving abilities and analytical thinking. Overall, this internship has played an important role in strengthening my foundation in cloud computing and preparing me for future opportunities in the AWS domain. It has increased my confidence in working with cloud technologies and provided me with the necessary skills to pursue a career in this field.